

1.1 Classes of Sets

Def 1.1.1: A collection of sets $F \subset P(\Omega)$ is called an algebra if

- $\Omega \in F$
- $A \in F \rightarrow A^c \in F$
- $A, B \in F \rightarrow A \cup B \in F$ (Closure under pairwise unions)

Def 1.1.2: $F \subset P(\Omega)$ is called an σ -Algebra if it is an algebra and it satisfies

- $A_n \in F$ for $n \geq 1 \rightarrow \bigcup_{n \geq 1} A_n \in F$

Prop 1.1.1: $F \subset P(\Omega)$ is an σ -Algebra if and only if it is an algebra and it satisfies

- $A_n \in F, A_n \subset A_{n+1}$ for all $n \rightarrow \bigcup_{n \geq 1} A_n \in F$

Def 1.1.3: If A is a class of subsets of Ω , then the σ -Algebra generated by A , denoted by $\sigma\langle A \rangle$ is defined as:

- $\sigma\langle A \rangle = \bigcap_{F \in I(A)} F$

Where $I(A) = \{F : A \subset F \text{ and } F \text{ is a } \sigma\text{-Algebra on } \Omega\}$ is the collection of all σ -Algebras containing the class A

Def 1.1.4: The Borel σ -Algebra on a topological space S (in particular, on a metric space or an Euclidean space) is defined as the σ -algebra generated by the collection of open sets in S .

Def 1.1.5: A class C of subsets of Ω is a π -system or a π -class if

- $A, B \in C \rightarrow A \cap B \in C$

Def 1.1.6: A class L of subset of Ω is a λ -system or λ -class if

- $\Omega \in L$
- $A, B \in L, A \subset B \rightarrow B \setminus A \in L$
- $A_n \in L, A_n \subset A_{n+1}$ for all $n \geq 1 \rightarrow \bigcap_{n \geq 1} A_n \in L$

Theorem 1.1.2: If C is a π -system then $\lambda\langle C \rangle = \sigma\langle C \rangle$

Corollary 1.1.3: If C is a π -system and L is a λ -system containing C , then $\sigma\langle C \rangle \subset L$:

1.2 Measures

Def 1.2.1: Let Ω be a nonempty set and F be an algebra on Ω . Then a set function μ on F is called a measure if

- $\mu(A) \in [0, \infty] \forall A \in F$
- $\mu(\emptyset) = 0$
- for any disjoint collect of sets $A_1, A_2, \dots \in F$ with $\bigcup_{n \geq 1} A_n \in F$,

$$\mu\left(\bigcup_{n \geq 1} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n)$$

Prop 1.2.1: Let Ω be a non-empty set and F be an algebra of subsets of Ω and μ be a set function on F with values in $[0, \infty]$ and with $\mu(\emptyset) = 0$. Then μ is a measure iff μ satisfies

- (finite additivity) for all $A_1, A_2 \in F$ with $A_1 \cap A_2 = \emptyset$, $\mu(A_1 \cup A_2) = \mu(A_1) + \mu(A_2)$
- (monotone continuity from below, mcfb) for any collection $\{A_n\}_{n \geq 1}$ of sets in F such that $A_n \subset A_{n+1}$ for all $n \geq 1$ and $\bigcup_{n \geq 1} A_n \in F$

$$\mu\left(\bigcup_{n \geq 1} A_n\right) = \lim_{n \rightarrow \infty} \mu(A_n)$$

Def 1.2.2: finite or infinite A measure μ is called finite or infinite according as $\mu(\Omega) < \infty$ or $\mu(\Omega) = \infty$. A finite measure with $\mu(\Omega) = 1$ is called a probability measure. A measure μ on a σ -algebra G is called σ -finite if there exist a countable collection of sets $A_1, A_2, \dots \in F$ not necessarily disjoint, such that

- $\bigcup_{n \geq 1} A_n = \Omega$
- $\mu(A_n) < \infty$ for all $n \geq 1$

Prop 1.2.2 Let μ be a measure on an algebra F and let $A, B, A_1, \dots, A_k \in F, 1 \leq k < \infty$. Then:

- (Monotonicity) $\mu(A) \leq \mu(B)$ if $A \subset B$
 - (Finite Subadditivity) $\mu(A_1 \cup \dots \cup A_k) \leq \mu(A_1) + \dots + \mu(A_k)$
 - (Inclusion-Exclusion Formula) if $\mu(A_i) < \infty$ for all $i = 1, \dots, k$ then,
 - $\mu(A_1 \cup \dots \cup A_k) = \sum_{i=1}^k \mu(A_i) - \sum_{1 \leq i < j \leq k} \mu(A_i \cap A_j) + \dots + (-1)^{k-1} \mu(A_1 \cap \dots \cap A_k)$
- Prop 1.2.3** let μ be a measure on an algebra F
- (Monotone Continuity from above) Let $\{A_n\}_{n \geq 1}$ be a sequence of sets in F such that $A_{n+1} \subset A_n$ for all $n \geq 1$ and $A = \bigcap_{n \geq 1} A_n \in F$. Also, let $\mu(A_{n_0}) < \infty$ for some $n_0 \in \mathbb{N}$. Then $\lim_{n \rightarrow \infty} \mu(A_n) = \mu(A)$
 - (countable subadditivity) if $\{A_n\}_{n \geq 1}$ is a sequence of sets in F such that $\bigcup_{n \geq 1} A_n \in F$, then

$$\mu\left(\bigcup_{n \geq 1} A_n\right) \leq \sum_{n=1}^{\infty} \mu(A_n)$$

Theorem 1.2.4: Uniqueness of Measures Let μ_1 and μ_2 be two finite measures on a measurable space (Ω, F) . Let $C \subset F$ be a π -system such that $F = \sigma\langle C \rangle$. if $\mu_1(C) = \mu_2(C)$ for all $C \in C$ and $\mu_1(\Omega) = \mu_2(\Omega)$, then $\mu_1(A) = \mu_2(A)$ for all $A \in F$

1.3 The extension theorems and Lebesgue-Stieltjes measures

1.3.1 Caratheodory Extension of Measures

Def 1.3.1: Let Ω be a nonempty set and let $P(\Omega)$ be the power set of Ω . A class $C \subset P(\Omega)$ is called a semialgebra if

- $A, B \in C \rightarrow A \cap B \in C$
- For any $A \in C$ there exists $B_1, \dots, B_k \in C$ for some $1 \leq k < \infty$ such that $B_i \cap B_j = \emptyset$ for $i \neq j$, and $A^c = \bigcup_{i=1}^k B_i$

Def 1.3.2: A set function μ on a Semialgebra $\mathcal{C}$, taking values in $\overline{\mathbb{R}}_+ = [0, \infty]$ is called a measure if

- $\mu(\emptyset) = 0$
- for any sequence of sets $\{A_n\}_{n \geq 1} \subset \mathcal{C}$ with $\bigcup_{n \geq 1} A_n \in \mathcal{C}$ and $A_i \cap A_j = \emptyset$ for $i \neq j$.
 $\mu(\bigcup_{n \geq 1} A_n) = \sum_{n=1}^{\infty} \mu(A_n)$

Def 1.3.3: Given a measure μ on a semialgebra $\mathcal{C}$, the outer-measure induced by μ is the set function μ^* , defined on $P(\Omega)$, as

$$\mu^*(A) = \inf \left\{ \sum_{n=1}^{\infty} \mu(A_n) : \{A_n\}_{n \geq 1} \subset \mathcal{C}, A \subset \bigcup_{n \geq 1} A_n \right\}$$

Def 1.3.4: A set A is said to be μ^* -measureable if

$$\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \cap A^c) \forall E \subset \Omega$$

Def 1.3.5: Any set function $\mu^* : P(\Omega) \rightarrow \overline{\mathbb{R}}_+ = [0, \infty]$ is an outer measure on Ω if it satisfies:

- $\mu^*(\emptyset) = 0$
- $A \subset B \rightarrow \mu^*(A) \leq \mu^*(B)$
- for any $\{A_n\}_{n \geq 1} \subset P(\Omega)$

$$\mu^*\left(\bigcup_{n \geq 1} A_n\right) \leq \sum_{n=1}^{\infty} \mu^*(A_n)$$

Theorem 1.3.2 Let μ^* be an outer measure on Ω . Let $\mathcal{M} = \mathcal{M}_{\mu^*} = \{A : A \text{ is } \mu^*\text{-measureable}\}$ Then,

- $\mathcal{M}$ is a σ -algebra
- μ^* is restricted to $\mathcal{M}$ is a measure
- $\mu^*(A) = 0 \rightarrow P(A) \subset \mathcal{M}$

Def 1.3.6 A measure space (Ω, F, v) is called complete if for any $A \in F$ with $v(A) = 0 \rightarrow P(A) \subset F$

Theorem 1.3.3 (Caratheodory's Extension Theorem). Let μ be a measure on a semialgebra $\mathcal{C}$ and let μ^* be an out measure as defined in definition 1.3.3, then:

- μ^* is an outer measure

- $\mathcal{C} \subset \mathcal{M}_{\mu^*}$
- $\mu^* = \mu$ on $\mathcal{C}$, where $\mathcal{M}_{\mu^*}$ is as in Theorem 1.3.2

Theorem 1.3.6 (Uniqueness). Let μ be a σ -finite measure on a semialgebra $\mathcal{C}$. Let v be a measure on the measureable space $(\Omega, \sigma(\mathcal{C}))$ such that $v = \mu$ on $\mathcal{C}$. Then, $v = \mu^*$ on $\sigma(\mathcal{C})$

Lebesgue-Stieltjes Measures on $\mathbb{R}$

Def 1.3.7 Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be nondecreasing. the measure space $(\mathbb{R}, \mathcal{M}_{\mu_F^*}, \mu_F^*)$ is called the Lebesgue-Stieltjes measure space and μ_F^* is the Lebesgue-Stieltjes measure generated by F

Def 1.3.8 (Lebesgue Measure on $\mathbb{R}$). When $F(x) = x, x \in \mathbb{R}$, the measure μ_F^* is called the Lebesgue Measure and the σ -algebra $\mathcal{M}_{\mu_F^*}$ is called the class of Lebesgue Measureable sets.